Basic Concepts

Computer Organization and Assembly Language

Lec#1

Computer Science Department

National University of Computer and Emerging Sciences Islamabad

Overview

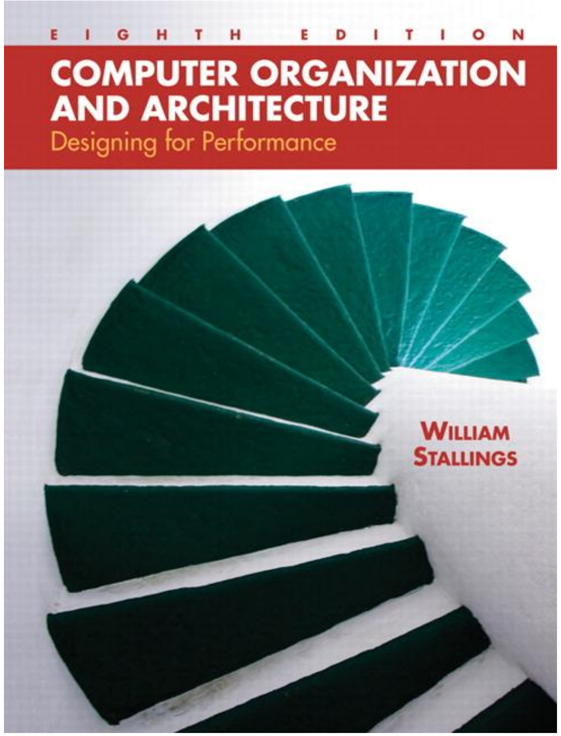
- Welcome to COAL 2024
- Course Contents
- Assembly-, Machine-, and High-Level Languages
- Assembly Language Programming Tools
- Programmer's View of a Computer System
- Data Representation

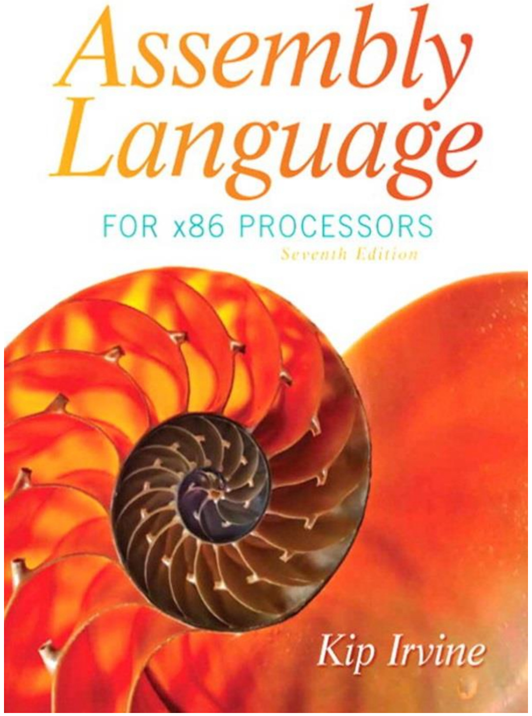
Class Particulars

- Credit Hours:
  - 4.0 [3+1]
- Course Structure:
  - Two Lectures a week (Each of duration 1.5 hour).
  - One lab of 3-hr in each week

Text Books

- Computer Organization and Architecture, William Stallings,
  - 10th edition
- Kip Irvine: Assembly Language for Intel-Based Computers
  - 7th edition
- Read the textbook!
  - Key for learning and obtaining a good grade





Grading Policy

- There is simply no chance of extension in any of the deadline, what so ever
- You can request for re-checking of any of your evaluation as per following rules;

  Exams: Same day

  Assignments: 2 days after handing-over

  Quizzes: 2 days after handing over

  Warning! After due time, request will not be considered even if it’s genuine.

  Warning! Keep checking your flex regularly and don’t come in the end with bulk of queries in hand which you never presented earlier and now you are on the Edge.

General Guidelines

- Visit GCR regularly for updates
- No email submissions when GCR is there.
- Cheating cases are intolerable. You will be given negative marks for cheated stuff irrespective of the fact that you were provider or the other one. Your cheating in exam will make it easy for you to step down from the Course with an ‘F’ grade. . . ☹
- There will be no re-take of any evaluation except for exams for which you should follow proper procedure
- Quiz is inevitable so always, expect One ☺

General Guidelines

- You have to depend on yourself to have a good grade in the course
  - *You need to appear in demo to have it graded otherwise you will be given 5% of the total marks even if it was best assignment/project of the class*
  - *Grading will be individual even for group tasks. So better to shine with your own work in hand otherwise don't complaint*
  - *There will always be a quiz of written assignments whose performance will be considered as performance in that assignment.*

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Course Contents- Organization

- Introduction to Computer Organization and Assembly Language
- Instruction Set Architecture
- Computer Evolution and Performance
- Computer Components
- A Top Level View of Computer Function
- Interconnection Structure
- Computer Memory System

Course Contents- Organization

- Performance of a Computer System
- Introduction to Microcontroller
- Microcontroller Vs Microprocessor
- Arduino Microcontroller
- Data Transfer and Addressing Modes
- Input Output Techniques
- Computer Arithmetic
- High Level Interfacing

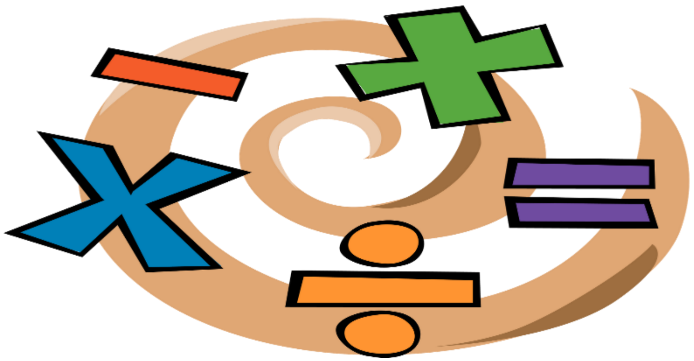
Course Contents- Assembly Language

- Basic Elements of Assembly Language
- Flat Memory Program Template
- Data-Related Operators & Directives
- Array Processing
- String Processing
- Stack Operations and Procedures

Computer Architecture

Architecture is those attributes visible to the programmer

- Instruction set, number of bits used for data representation, I/O mechanisms, addressing techniques.
- e.g. Is there a multiply instruction?



4 Bits	6 Bits	6 Bits
Opcode	Operand reference	Operand reference

← 16 Bits →

A Simple Instruction Format

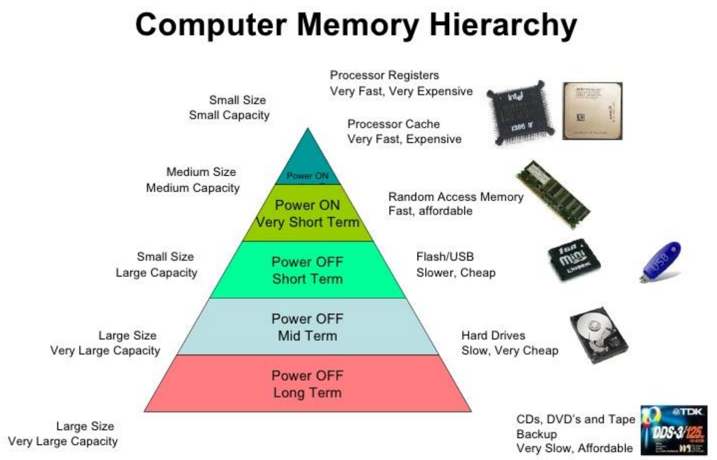
Computer Organization

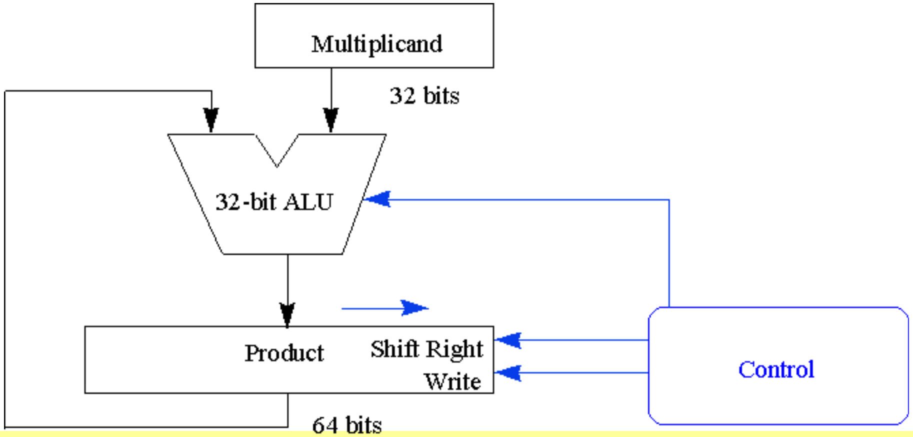
Organization is how features are implemented

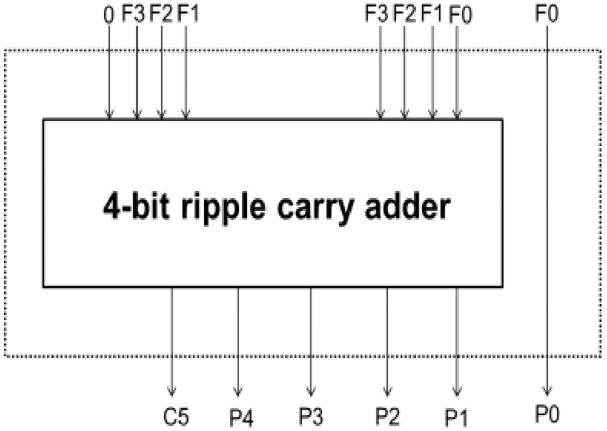
- Control signals, interfaces, memory technology.

- how different components of computer are linked together to meet the requirements

- e.g. Is there a hardware multiply unit or is it done by repeated addition?







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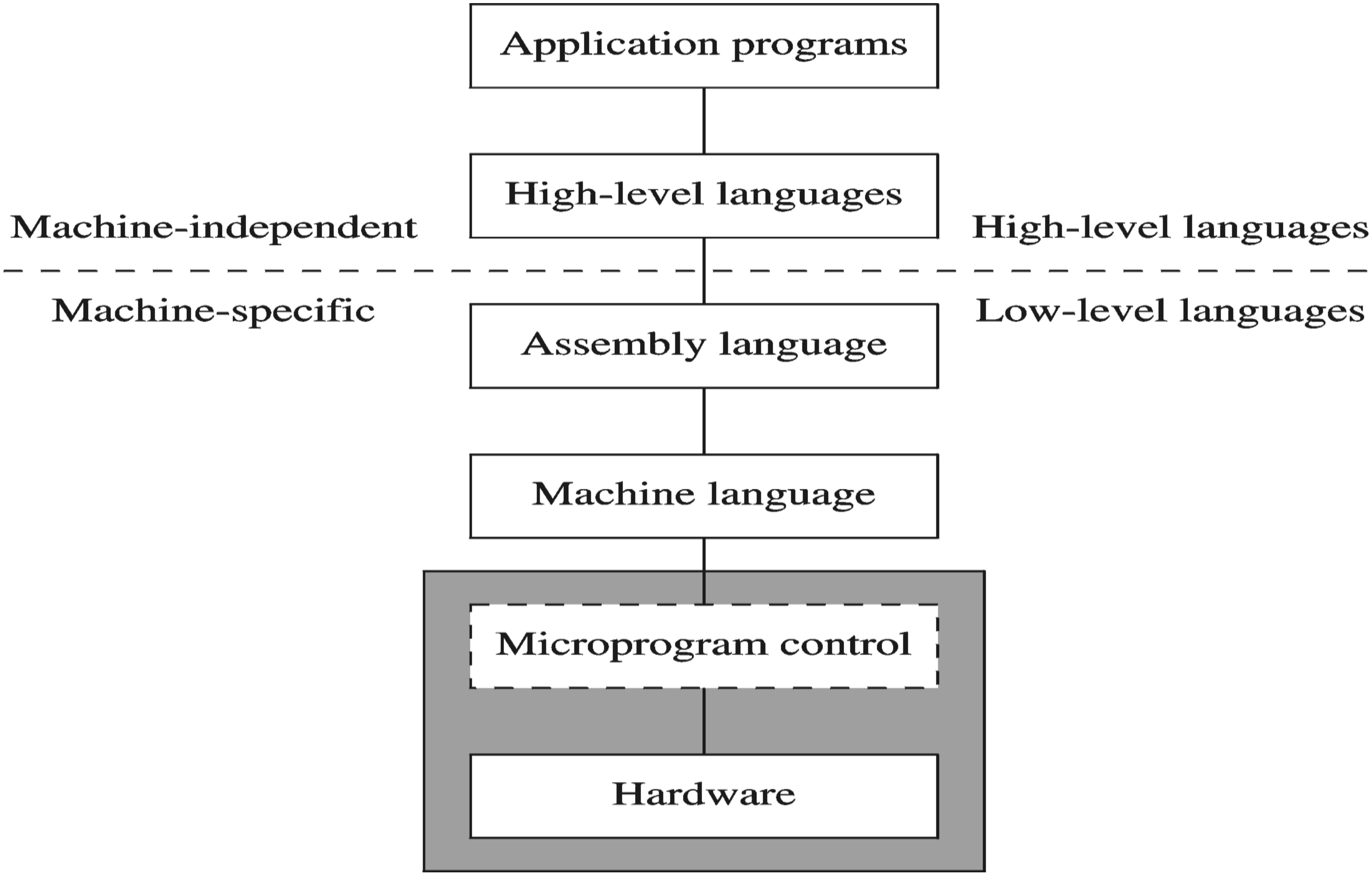
ASSEMBLY LANGUAGE

Some Important Questions to Ask

- What is Assembly Language?
- Why Learn Assembly Language?
- What is Machine Language?
- How is Assembly related to Machine Language?
- What is an Assembler?
- How is Assembly related to High-Level Language?
- Is Assembly Language portable?

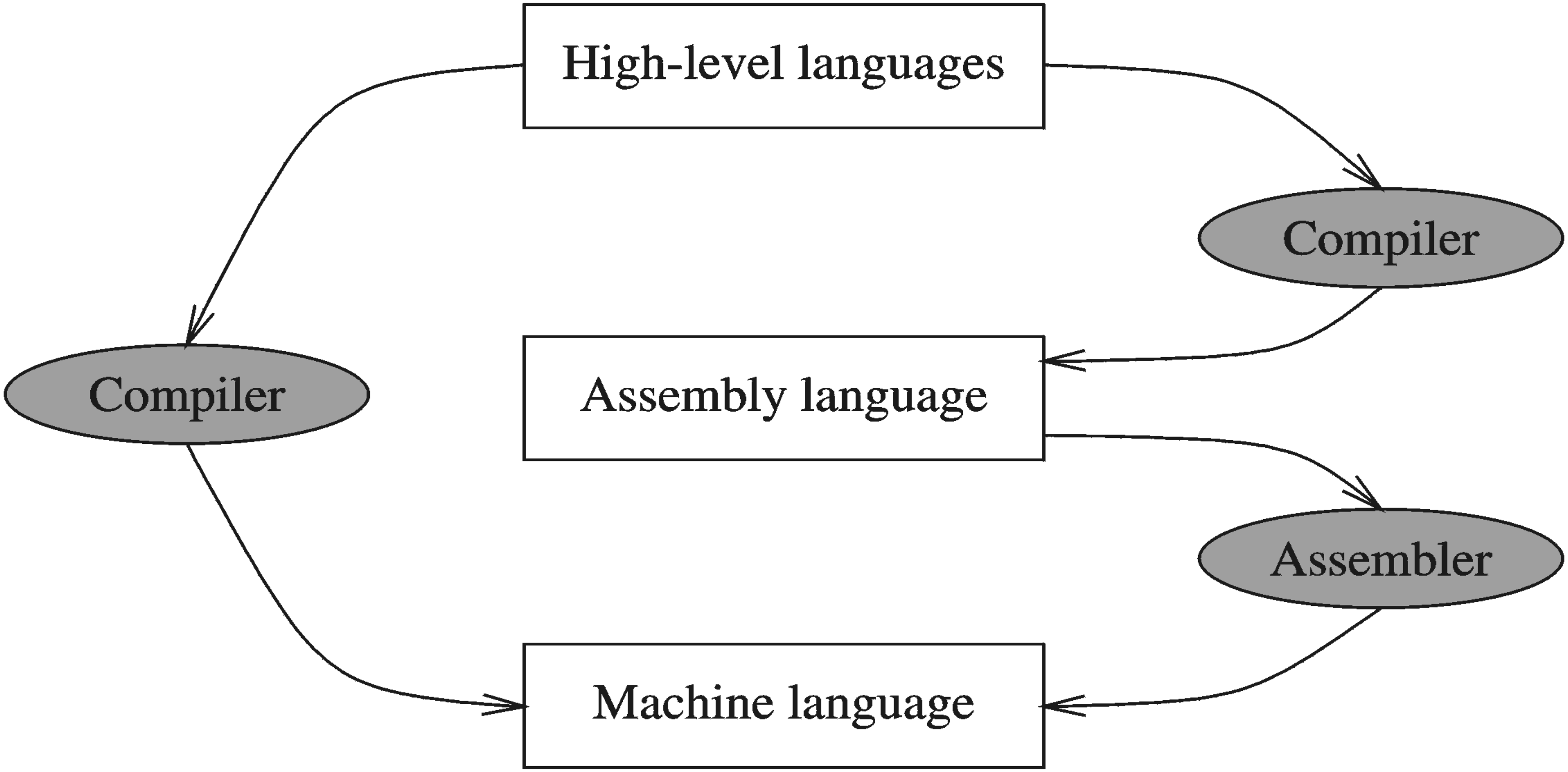
A Hierarchy of Languages



Assembly and Machine Language

- Machine language
  - Native to a processor: executed directly by hardware
  - Instructions consist of binary code: 1s and 0s
- Assembly language
  - Slightly higher-level language
  - Readability of instructions is better than machine language
  - One-to-one correspondence with machine language instructions
- Assemblers translate assembly to machine code
- Compilers translate high-level programs to machine code
  - Either directly, or
  - Indirectly via an assembler

Compiler and Assembler



Translating Languages

English: D is assigned the sum of A times B plus 10.

High-Level Language: D = A * B + 10

A statement in a high-level language is translated typically into several machine-level instructions

Intel Assembly Language:	Intel Machine Language:
mov eax, A	A1 00404000
mul B	F7 25 00404004
add eax, 10	83 C0 0A
mov D, eax	A3 00404008

Advantages of High-Level Languages

- Program development is faster
  - High-level statements: fewer instructions to code
- Program maintenance is easier
  - For the same above reasons
- Programs are portable
  - Contain few machine-dependent details
    - Can be used with little or no modifications on different machines
  - Compiler translates to the target machine language
  - However, Assembly language programs are not portable

Why Learn Assembly Language?

- Two main reasons:
  - Accessibility to system hardware
  - Space and time efficiency
- Accessibility to system hardware
  - Assembly Language is useful for implementing system software
  - Also useful for small embedded system applications
- Space and Time efficiency
  - Understanding sources of program inefficiency
  - Tuning program performance
  - Writing compact code

Assembly vs High-Level Languages

- Some representative types of applications:

Type of Application	High-Level Languages	Assembly Language
Business application software, written for single platform, medium to large size.	Formal structures make it easy to organize and maintain large sections of code.	Minimal formal structure, so one must be imposed by programmers who have varying levels of experience. This leads to difficulties maintaining existing code.
Hardware device driver.	Language may not provide for direct hardware access. Even if it does, awkward coding techniques must often be used, resulting in maintenance difficulties.	Hardware access is straightforward and simple. Easy to maintain when programs are short and well documented.
Business application written for multiple platforms (different operating systems).	Usually very portable. The source code can be recompiled on each target operating system with minimal changes.	Must be recoded separately for each platform, often using an assembler with a different syntax. Difficult to maintain.
Embedded systems and computer games requiring direct hardware access.	Produces too much executable code, and may not run efficiently.	Ideal, because the executable code is small and runs quickly.

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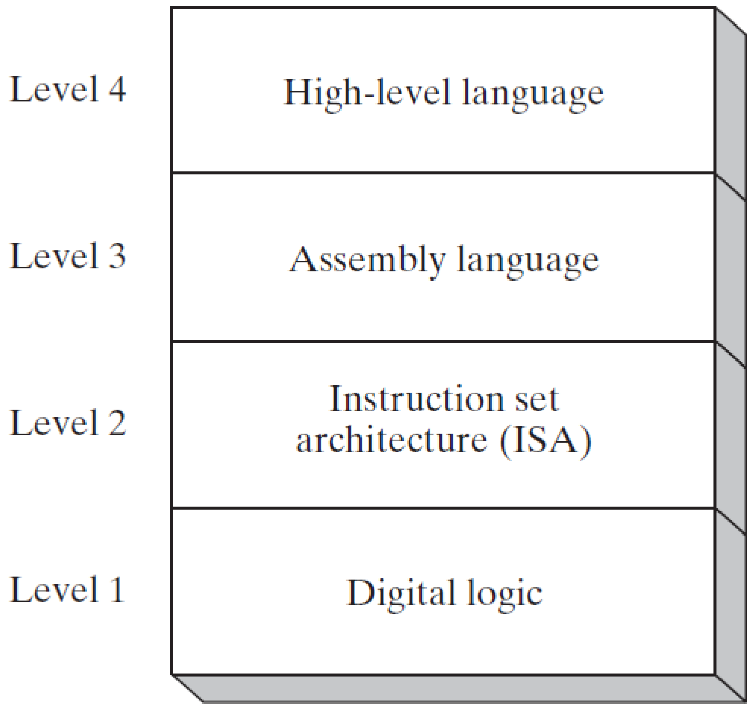
Programmer's View of a Computer System

Increased level of abstraction



Each level hides the details of the level below it

Programmer's View – 2

- **Application Programs (Level 4)**
  - Written in high-level programming languages
  - Such as Java, C++, Pascal, Visual Basic . . .
  - Programs compile into assembly language level (Level 4)

- **Assembly Language (Level 3)**
  - Instruction mnemonics are used
  - Have one-to-one correspondence to machine language
  - Calls functions written at the operating system level
  - Programs are translated into machine language

Programmer's View – 3

- **Instruction Set Architecture (Level 2)**
  - Specifies how a processor functions
  - Machine instructions, registers, and memory are exposed
  - Machine language is executed by Level 1 (microarchitecture)

- **Digital Logic (Level 1)**
  - Implements the microarchitecture
  - Uses digital logic gates
  - Logic gates are implemented using transistors

Reference

- Chapter #1
  - Assembly Language for x86 Processor 7th Edition
    - By Kip Irvine

THAT'S IT FOR TODAY